A. Ebrar Kızıloğlu

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Research interests: Multilingual & low-resource NLP, morphology-informed language modeling, cross-lingual representation transfer, and evaluation methodology.

Education

- **Technical University of Munich (TUM)** 2025 – 2027
  MSc in Informatics *GPA: 1.60*
  - Machine Learning Track
  - **Selected Coursework**: Advanced Natural Language Processing, Introduction to Deep Learning, Protein Prediction
- **Bogazici University** 2020 – 2025
  BSc in Computer Engineering *GPA: 3.44*

Publications

- M. Türker, **A. E. Kızıloğlu**, O. Güngör, S. Üsküdarlı, *TabiBERT: A Large-Scale ModernBERT Foundation Model and a Unified Benchmark for Turkish Language*, arXiv preprint arXiv:2512.23065, 2025. Under review at the Cambridge NLP Journal.
- M. Türker, **A. E. Kızıloğlu**, O. Güngör, S. Üsküdarlı, *TabiBERT: A Large-Scale ModernBERT Foundation Model and a Unified Benchmark for Turkish Language*, arXiv preprint arXiv:2512.23065, 2025. Under review at the Cambridge NLP Journal.

Experience

- **Co-supervisor, Bachelor Thesis** Sep 2026 – Present
  Bogazici University, Department of Computer Engineering *Istanbul, Türkiye*
  - Co-supervising a bachelor project on how well **language models generate correctly inflected Turkish words** (root + grammatical features → inflected form), covering suffix choice and order, vowel harmony and consonant alternations.
  - Guiding students to build a **benchmark of Turkish morphological generation tasks** (common and complex suffix combinations) using the **Zemberek** morphological analyzer, a reproducible evaluation program, a cross-model comparison, and a **taxonomy of generation errors**.
- **HiWi Researcher** Nov 2025 – Present
  TUM, Data Analytics & Statistics Lab *Munich, Germany*
  - Designed and ran a controlled **two-phase pretraining study** of whether **case-marking knowledge transfers from Turkish to German**: trained GPT-2-style **character-level LMs** on Turkish or English (control), then on German.
  - Built the data and evaluation pipeline: German **Accusative↔Dative minimal-pair** test sets derived from Universal Dependencies, and corpora subsampled from NLLB. Measured case distributions with Turkish and German morphological analyzers (Zemberek and SMOR).
  - Designed **ablations**: word shuffling, Turkish case-suffix stripping, and embedding resets.
  - Analyzed results with **perplexity-based probing**, multi-seed significance testing and **TOST equivalence tests**.
  - The primary findings suggest that the inclusion of Phase 1 pre-training improves German case accuracy by ∼4 points, but found no reliable Turkish advantage over English. Also, resetting the embeddings between the phases hurt English more than Turkish. Writing up the full findings as a paper (*in preparation*).
- **Research Assistant** Jan 2025 – Jan 2026
  Bogazici University, Tabilab *Istanbul, Türkiye*
  - Carried out the hands-on work to build a **modern encoder-only foundation model for Turkish**: data collection, pretraining, evaluation and open release, resulting in a co-authored paper (arXiv, under review).
  - Constructed a large-scale Turkish pretraining dataset of **84.88B tokens** from multiple corpora.
  - Pre-trained **TabiBERT**, an encoder-only model based on the ModernBERT architecture, for **1T** tokens.
  - Developed **TabiBench**, a unified benchmark for Turkish NLP, and compared TabiBERT against the major BERT-based Turkish models. **TabiBERT** established new **state-of-the-art** results on **5 of 8** tasks.
- **SWE Research Intern** Jul 2023 – Oct 2023
  Google, Text-to-Speech Research *London, UK*

- Tackled **robustness of the neural vocoder** (WaveGRU, a WaveNet-family model) to out-of-distribution acoustic features, e.g. squeaks when the VAE-based acoustic model, prosody control or voice conversion requests an extreme pitch.
- Designed an **F0-aware loss** with an auxiliary training path: artificial features sampled far from the mean of the acoustic model's VAE (σ=5) are vocoded, F0 is re-extracted with a **differentiable** estimator (FiPPiE), and a **log-F0 MSE** loss is added to the standard loss. No architecture change.
- Cut **Gross Pitch Error** on outlier F0 from 4.64% to 1.66% (**64%** relative), improving audio quality for 90M+ Assistant users. Analyzed the loss-weight trade-off and the quality/F0-fidelity trade-off with **MOS and AB tests**.

- **STEP Intern** Jul 2022 – Oct 2022
  Google, Calendar *Zurich, Switzerland*
  - Worked on Calendar's **shared Java codebase**. Latency regressions in its hot loops directly affect user-visible latency, so the team needed a **write-once, run-everywhere benchmarking tool**.
  - Surveyed the benchmarking and testing frameworks used at Google, and settled on **Caliper** for Android.
  - Built benchmarks for the performance-critical event-fetching method at four data densities (empty, minimal input with 100% line coverage, thousands of generated events, and real calendar data).
  - Built the pipeline to benchmark on **real-world data**: a **PII scrubber** for exported calendars, a generator of Java builder classes that reproduce the scrubbed data, and a shell script automating the steps.
  - Analyzed latency against data size and presented the results to the Calendar Android, iOS and Web teams.

Presentations

- **LatentShift Conference** 2026
  Invited talk on **TabiBERT** at LatentShift, a conference for engineers who build, ship, and run AI in production. Covered the full research pipeline: building the 84.88B-token Turkish corpus, ModernBERT-based pretraining, the TabiBench evaluation suite with state-of-the-art results on 5 of 8 tasks, and the practical advantages of a modern encoder for production Turkish NLP.

Projects

- **LLM-driven Turing Game Bot (Bachelor Thesis),** 2024
  Developed a **GPT-4o chatbot** achieving **31% human deception** in Turing Game sessions through **120+ user tests** with 24 participants. Benchmarked against Mistral, LLaMA, and GPT, optimizing prompts for higher deception rates.
  Implemented with **Python, Node.js, SQLite, Docker**, and **Nginx** for scalable deployment.

Achievements

IELTS Overall Band Score: **8.5** 2025
TUBITAK Scholarship recipient, awarded for medals in national and international mathematical olympiads. 2020 – 2024
Silver Medalist, European Girls' Mathematical Olympiad (EGMO), representing Türkiye; proof-based competition against national teams from 50+ countries. 2019
Silver Medalist, Turkish National Mathematical Olympiad (UMO), the selection route to the national team. 2019